

EVAPORATION OF PURE LIQUIDS FROM OPEN SURFACES

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Abstract

We developed an estimation technique for the evaporation rate of volatile liquids from open surfaces in the presence of a flowing air stream which, we believe, is an improvement over previous methods, particularly when the air velocity is greater than about 50 centimeters per second (100 feet per minute). The estimation technique is predicated on solution of the classical mass transfer coefficient model with the gas-phase mass transfer coefficient estimated by the correlation of Mackay and Matsugu. The new technique requires only the vapor pressure and molecular weight of the vaporizing substance, the ambient air temperature and velocity, and the area of the liquid pool for its solution. Other properties of the ambient air are incorporated into the model by the kinetic theory of gases. Our model was tested against experimental results on 19 different organic compounds generated by four different experimenters over a wide range of experimental conditions. The model matched the experimental data well, usually within 20 percent. The model can be used to obtain estimates of the evaporation rates of organic liquids resulting from spills or open surface operations.

Introduction

When reviewing premanufacture notifications for the production of new chemical substances as mandated by the Toxic Substances Control Act, the U.S. Environmental Protection Agency must evaluate potential exposures of workers to new chemicals during their manufacture, processing, and use. Therefore, the Agency is interested in developing methods for predicting worker exposures from the minimal amount of physical property data available on new chemical substances. For those chemicals which are sufficiently volatile to result in exposure by inhalation, the rate of evaporation into the ambient atmosphere must be predicted with reasonable accuracy.

To calculate the rate of evaporation of a liquid, not only the physical properties of the liquid, but also the properties of the ambient atmosphere such as the air temperature,

pressure, and velocity must be known. Precise calculation of evaporation rates requires a detailed understanding of the energy transfer between the volatile liquid, the air, and any solid surfaces in contact with the liquid. It is virtually impossible to know or be able to measure all these conditions. However, if some simplifying assumptions can be made with regard to this complex system, it is possible to arrive at estimates of vaporization rates for a variety of volatile substances that are, in most cases, within twenty percent of experimental results.

The intent of this work was to develop a model of pure liquid evaporation from open surfaces based on the mass transfer coefficient correlation of Mackay and Matsugu (1973) and compare the model to penetration and mass transfer coefficient models from the literature. The assumptions made in developing these models are the same. They include the following: the contaminant is emitted from a single source; the surface area of the liquid pool does not change appreciably with time; air velocity above the pool is constant; there is no back pressure from the vaporized chemical which retards the rate of evaporation; and, the ideal gas law applies to all constituents. The resulting models require only the molecular weight and vapor pressure of the contaminant, ambient pressure and temperature, liquid pool size, and local air velocity for their solution.

Derivation of the mass transfer coefficient model incorporating the correlation of Mackay and Matsugu is described in the following section. This is followed by a description of the penetration model and mass transfer coefficient models from the literature. Subsequent sections compare the predictions of evaporation rates using the models to experimental data.

The Mass Transfer Coefficient Model

Colburn and Drew (1937) analogized mass transfer to heat transfer which resulted in an equation for mass transfer across an interface:

$$N_A = K \frac{P_v}{RT} \quad (1)$$

where N_A = mass transfer rate, g-moles/cm²-sec
 K = mass transfer coefficient, cm/sec
 P_v = vapor pressure of the evaporating liquid, kpa
 R = universal gas constant, 8.314 x 10³ cm³-kpa/gm-mole-°K
 T = absolute temperature, °K.

Mackay and Matsugu (1973) used data on the evaporation of water and cumene from shallow open pans to correlate the gas-phase mass transfer coefficient as a function of wind-speed and liquid pool size by the equation

$$K = 0.0220 v_z^{0.78} \Delta z^{-0.11} Sc^{-0.67} \quad (2)$$

where K = mass transfer coefficient, cm/s
 v_z = wind speed, cm/s
 Δz = length of the air-liquid interface in the direction of flow, cm
 Sc = Schmidt number, the dimensionless ratio of air viscosity divided by the product of air density and diffusivity of the evaporating species, i.e. $Sc = \mu/\rho D_{AB}$.

The exponents on the velocity and length terms result from the Von Karman similarity principle as modified by Sutton (1953) and are reported to have provided satisfactory results when the air flow is in or near the turbulent range. The Schmidt number is used to correlate mass and momentum transfer between dynamically similar systems. The transfer rate can be converted to mass units by the relationship

$$G = N_A M_A \quad (3)$$

where G = liquid evaporation rate, gm/sec/cm²
 M_A = molecular weight, gm/gm-mole.

Combining Equations 1, 2, and 3, we obtain

$$G = 2.65 \times 10^{-6} \frac{M_A p v_z^{0.78}}{\Delta z^{0.11} Sc^{0.67} T}$$

The Chapman-Enskog Theory can be used to estimate the viscosity of a gas, in this case, air. Details of the approximation are given by Bird et al. (1960). Over the temperature range from 10 to 70 °C, the viscosity may be approximated by

$$\mu = -9.426 \times 10^{-5} + 1.610 \times 10^{-5} \sqrt{T} \quad (5)$$

where μ = viscosity of air at the ambient temperature, gm/cm/sec
 T = absolute temperature, °K.

Similarly, the Chapman-Enskog Theory can be used to estimate the diffusion coefficient, D_{AB} , if the absolute temperature, pressure, and the molecular weight of the evaporating species are known.

$$D_{AB} = \frac{4.09 \times 10^{-5} T^{1.9} \sqrt{\frac{1}{28.97} + \frac{1}{M_A}}}{P M_A^{0.33}} \quad (6)$$

where D_{AB} = diffusivity of species A in air, cm/sec
 P = ambient air pressure, atmospheres.

Finally, at pressures below 50 psia, the density of air may be determined from the ideal gas law.

$$\rho = 0.352 \frac{P}{T} \quad (7)$$

where ρ = density of ambient air, gm/cm³
 T = ambient air temperature, °K.

Combining Equations 5, 6, and 7, an expression for the Schmidt number is obtained.

$$Sc = \frac{\mu}{\rho D_{AB}} = \frac{(-6.57 + 1.12\sqrt{T})M_A^{0.33}}{T^{0.9} \sqrt{\frac{1}{28.97} + \frac{1}{M_A}}} \quad (8)$$

Although the Schmidt number is constrained by the foregoing assumptions, for general environmental assumptions, its value lies between one and two. Therefore, its impact on the mass transfer coefficient is relatively small. Equation 7 for the Schmidt number may be substituted into Equation 3a for the evaporation rate to yield a single equation for calculation of the rate of evaporation of an organic liquid from a quiescent pool.

$$G = 4.71 \times 10^{-6} \frac{M_A^{0.78} p_v v_z^{0.78} (1/28.97 + 1/M_A)^{0.33}}{\Delta z^{0.11} T^{0.4} (-5.87 + \sqrt{T})^{2/3}} \quad (9)$$

Alternate Models from the Literature

Hummel, Braun, and Fehrenbacher (1996) developed a penetration model for diffusion of a evaporating liquid into a flowing gas using Fick's second law of diffusion. In terms of the variables as defined above, their expression for the rate of evaporation is given by

$$G = 8.68 \times 10^{-7} \frac{M_A^{0.833} p_v v_z^{0.5} (1/28.97 + 1/M_A)^{0.25}}{\Delta z^{0.5} T^{0.05}} \quad (10)$$

Lennert, Nielsen, and Breum (1997) at the Danish National Institute of Occupational Health studied the performance of seven correlations for predicting evaporation rates. The models applicable to turbulent flow conditions (Gmehling et al., 1989; Gray, 1974; and, Olsen et al., 1992) were tested in this study.

Evaluation of the Mass Transfer Coefficient Model

Several sources of experimental data were used to evaluate the validity of the mass transfer coefficient model. The single largest source of experimental measurements is a study by Pace Laboratories funded by the U.S. Environmental Protection Agency (1992). Additional data were obtained from the papers of Wade (1942), O'Brien and Stutzman (1950), and Gray (1974). All expressions for the rate of evaporation of a volatile liquid listed herein were compared to these experimental data.

Pace Data

In this study, 14 organic liquids¹ were evaporated from a small, temperature controlled pan set in a rectangular duct into an air stream passing over the pan.² More than 150 test runs were conducted over temperatures ranging from 40°F to 120°F and air flow rates between 100 and 1500 feet per minute. The predictions of the Penetration Model developed by Hummel et al. (1996) and the Mass Transfer Coefficient Model developed herein are compared to the measured evaporation rates in Figure 1. Omitted from this graph are the measurements at 100 feet per minute; at this velocity, the air flow is in transition from laminar to turbulent flow and the mass transfer coefficient model is inapplicable because it assumes fully developed turbulent flow. If the prediction methods were to match the experimental data perfectly, all of the data points would lie on the unit slope line on the graph. As can be seen, for turbulent flow conditions, the penetration model significantly over-predicts the evaporation rate; the average error is +30% and the root mean square error (the square root of the mean square error) is 49%. The mass transfer coefficient model under-predicts the experimental data slightly; the average error is -10% while the root mean square error is 32%.

The correlations applicable to turbulent flow studied by Lennart et al. (1997) were also compared to the experimental Pace data. Of these models, the NiOH-2 model provided the best match with an average error of -20% and a root mean square error of 32%. The average errors of the BAU, SUBTEC, and Gray models were all near 70%.

Data of Wade

In 1942, S. H. Wade reported his experiments on the rate of evaporation of six low-boiling

¹ Hexane, n-heptane, octane, benzene, toluene, xylene, acetone, methyl ethyl ketone, octanone, methanol, propanol, pentanol 1-hexanol, heptanol, and 2-octanol

² Although the Pace experiments included vaporization of water, the models discussed in this article are not recommended to predict evaporation rates for water.

liquids³ from a shallow tray within a wind tunnel. His data was presented in graphical form, showing the measured evaporation rate as a function of vapor pressure with air velocity over the liquid pool as the parametric variable. Wade reported evaporation rates up to 0.15 grams per second and air velocities as great as 393 cm/sec (774 ft/min). The wind speed, liquid vapor pressure, and evaporation rate were read from the graphs and the liquid temperature inferred from the vapor pressure of the pure liquid. With this information, the predicted rate of evaporation could be calculated and compared to the experimental value.

Figure 2 compares the model predictions to the rates of evaporation measured by Wade. In general, the evaporation rates predicted with the Mass Transport Coefficient Model are superior to those predicted by the Penetration Model. The average error was -14% and the root mean square error was 18% using the former while the average error was 87% and the root mean square error was 95 % using the latter. The NIOH-2 model of Lennart et al. (1997) provided estimates of the evaporation rate equivalent to the Mass Transfer Coefficient model but with a larger root mean square error.

Data of O'Brien and Stutzman

L. J. O'Brien and L.F. Stutzman (1950) published a paper in 1950 in which they describe the evaporation of four volatile organic liquids in an apparatus similar to that used by Pace Laboratories and by Wade. In all of their runs, the air was heated to 100°F while, in most cases, the liquid was also heated. Under these conditions, vapor pressures ranging from 22 to over 100 torr were found to occur. Air velocities ranged from 59 to almost 500 feet per minute.

The results of applying the two models are shown in Figure 3. The Mass Transport Coefficient Model matches the data adequately with an average error of 16% and a root mean square error of 18%. The Penetration Model predicted the evaporation rates poorly. The average error of this model was 111% and the root mean square error was 117%. For the evaporation rates in this study, predictions made with the Mass Transfer Coefficient Model was superior for laminar as well as turbulent flow conditions. Of the models studied by Lennart and coworkers, the NIOH-2 model was equivalent to the Mass Transfer Coefficient model.

Data of Gray

Douglas Gray reported data from the vaporization of xylene in a square wind tunnel, 6 inches on a side. The sampling well was 6 inches square and 1/4-inch deep. Air and liquid temperatures were between 70 and 77°F in all experiments. Air velocity ranged from 11.9 cm/sec (23.4 ft/min) to 376 cm/sec (740 ft/min). The measured and predicted evaporation rates are compared in Figure 4. The Mass Transfer Coefficient Model tends to under-

³ Acetone, benzene, carbon tetrachloride, ethyl acetate, toluene, and trichloroethylene

predict the evaporation rate for laminar flow conditions (when it is inappropriate) and to over-predict the evaporation rate when the air flow is turbulent. The average error of all predictions is -11% and the root mean square error is 17%. Predictions of evaporation rates made using the Penetration Rate Model are uniformly high; the average overestimation is 35% and the root mean square error is 42%. Of the models studied by Lennart and coworkers, the NIOH-2 model was comparable to the Mass Transfer Coefficient Model; the other models were less satisfactory.

Comparison at Laminar Flow Conditions

As noted previously, the comparison of predictive models for evaporation rates in this paper was intentionally limited to turbulent flow conditions at which the model of Mackay and Matsugu (1973) was developed. However, a survey by Baldwin and Maynard (1998) of the wind speed in 55 work areas covering a wide range of workplaces produced an arithmetic mean speed of 0.3 meters per second. The measurements were highly skewed with 85% of the individual measurements below the mean value. Accordingly, air flow in many indoor environments is laminar.

The Pace data taken at a wind speed of 100 feet per minute (50 cm/sec) is in the transition region from laminar to turbulent. At this wind speed, all of the models considered in this paper predicted an evaporation rate of the volatile liquid that was less than the measured rate. The Penetration Model provided the best estimates at this wind speed with an average underestimation of 20%. Predictions of evaporation rates made with the Mass Transfer Coefficient Model and the other models were, on average, less than half the measured evaporation rate.

Conclusion

Both the Mass Transfer Coefficient and the Penetration Models are completely general and can be applied to predict the vaporization of any pure liquid for which the molecular weight, vapor pressure, and liquid temperature are known. Experience suggests that the vapor pressure should be measured or calculated by a suitable correlation at the liquid temperature.

Before judging the validity of the models, some discussion of the data available for evaluation of the models is in order. First, the experimental design used by Pace Laboratories heated the liquid to minimize heat transfer effects on the rate of vaporization. It is doubtful if this aim was achieved. Second, their determination of the air flow regime, laminar or turbulent, was, at best, questionable given the design of the test section of their experimental apparatus.

Neither Wade, O'Brien and Stutzman, nor Grey made any effort to avoid heat transfer effects in their experiments. However, in practical situations compensation for heat transfer

effects is perhaps impossible and so we must settle for the liquid temperature as being the variable that can be measured and the best indicator of the vapor pressure near the air-liquid interface.

In light of the foregoing, care should be used in using the models described in this paper to predict evaporation rates. Nevertheless, the following generalizations can be made:

The Penetration Model

1. Is most appropriately used at very low air velocities (less than 50 centimeters per second), such as are encountered in unventilated interior environments, and
2. Should probably not be used when the vapor pressure of the evaporating liquid is in excess of 30 to 40 torr.

The Mass Transfer Coefficient Model

1. Will provide satisfactory estimates of liquid evaporation rates under virtually all conditions except when the air flow rate 50 centimeters per second or less, and,
2. Will yield acceptable estimates of vaporization at high vapor pressures and turbulent air flow conditions.

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